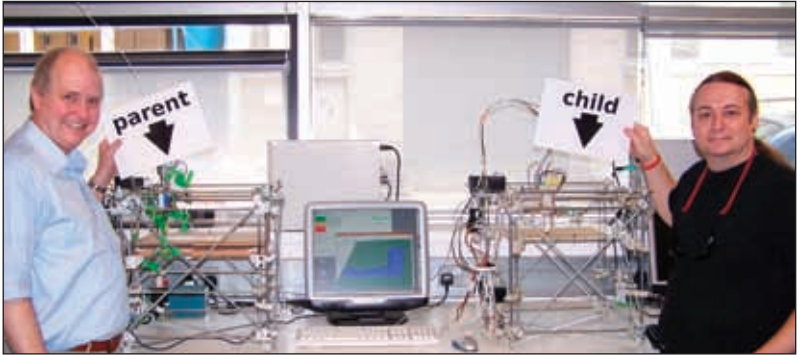




The project founder of RepRap Adrian Bowyer (left), and the software developer Vik Olver (right) showing a RepRap machine created by another RepRap machine.

software encourage you to find out how they work, and contribute back to the community.

We now see such a culture developing around hardware devices as well. While few will trust an open source hardware device designed by unknown community members, a free flow of information regarding hardware designs simply means that hardware will have the opportunity to innovate as fast as software. One example of a popular open source hardware platform is Arduino. It is a hardware platform which uses an Atmel ATmega168 microcontroller, and is designed to be simple to program and interface with a computer. Arduino also includes a software component – which is also open source – which is an IDE and programming interface for Arduino hardware. Since the specifications for Arduino are open and free, any hardware manufacturer can create Arduino-based boards which will work with the Arduino IDE. If you had any Arduino-based device, you could simply connect it to your computer and reprogram it to do whatever you wanted.

One of the most brilliant projects based on Arduino would have to be the RepRap project (reprap.org). The RepRap (Replicating Rapid-prototyper) device is a 3D printer that has an open design, and is developed with community participation. The firmware, and the actual design of the device are both open, so people are free to make their own RepRap machines and to improve upon them.

One of the most brilliant aspects of RepRap is that it can print many of its own parts! One you have one RepRap machine, you can construct another one using parts available in stores and parts printed with the RepRap machine! What better example could there be of a self-propagating open source hardware device?

While developing this device the engineers found that the specifications of the current Arduino-based boards were not enough for a project of this